Operations with radical expressions

1 Express the following as a single radical expression:

a $\sqrt{19} \times \sqrt{19}$

b $\sqrt{19} \times \sqrt{17}$

2 Multiply and simplify:

a $\left(\sqrt{45}\right)^2$

b $\sqrt{5} \times \sqrt{5}$

c $\sqrt{2} \times \sqrt{11}$

d $15 \times 4\sqrt{13}$

e $2\sqrt{2} \times 2$

f $5\sqrt{7} \times 3\sqrt{5}$

g $\sqrt{5} \times \sqrt{2} \times \sqrt{7}$

h $\sqrt{35} \times \sqrt{5}$

i $4\sqrt{35} \times \sqrt{5}$

j $4\sqrt{33} \times 4\sqrt{11}$

k $\sqrt{343} \times \sqrt{75}$

3 Consider the equation $\sqrt{k} + \sqrt{m} = \sqrt{k+m}$ and answer the following questions.

a Evaluate each of the following to one decimal place:

i $\sqrt{7}$

ii $\sqrt{20}$

iii $\sqrt{27}$

b Does $\sqrt{7} + \sqrt{20} = \sqrt{27}$?

c In general does $\sqrt{k} + \sqrt{m} = \sqrt{k+m}$?

4 Is the equation $2 - \sqrt{2} = \sqrt{2}$ true or false?

5 Simplify:

a $8\sqrt{3} + 18\sqrt{3}$

b $\sqrt{10} + 10\sqrt{10}$

c $\sqrt{3} - 16\sqrt{3}$

d $11\sqrt{7} - 5\sqrt{7}$

e $-19\sqrt{5} + 4\sqrt{5}$

f $-19\sqrt{10} - 15\sqrt{10}$

g $16\sqrt{3} + 3\sqrt{3} + 15\sqrt{3}$

h $20\sqrt{10} + 3\sqrt{10} - 6\sqrt{10}$

i $18\sqrt{2} - 10\sqrt{2} + 20\sqrt{2}$

j $-19\sqrt{2} + 2\sqrt{2} + 8\sqrt{2}$

k $6\sqrt{2} - 8\sqrt{2} - 17\sqrt{2}$

l $-12\sqrt{5} + 6\sqrt{5} - 12\sqrt{5}$

m $-16\sqrt{10} - 11\sqrt{10} + 18\sqrt{10}$

n $-11\sqrt{7} - 8\sqrt{7} - 2\sqrt{7}$

o $23\sqrt{3} + 26\sqrt{11} + 16\sqrt{3} + 19\sqrt{11}$

p $7\sqrt{3} + 7\sqrt{5} - 8\sqrt{3} + 10\sqrt{5}$

q $17\sqrt{6} - 8\sqrt{11} + 3\sqrt{6} + 13\sqrt{11}$

r $27\sqrt{2} + 24\sqrt{3} + \sqrt{2} + 14\sqrt{3}$

s $30\sqrt{13} + 26\sqrt{6} - 7\sqrt{13} - 16\sqrt{6}$

t $19\sqrt{3} + 19\sqrt{7} - 7\sqrt{7} + 22\sqrt{3}$

u $\dfrac{12 + 9\sqrt{5}}{3}$

v $\dfrac{4\sqrt{3}}{3} + \dfrac{\sqrt{3}}{6}$

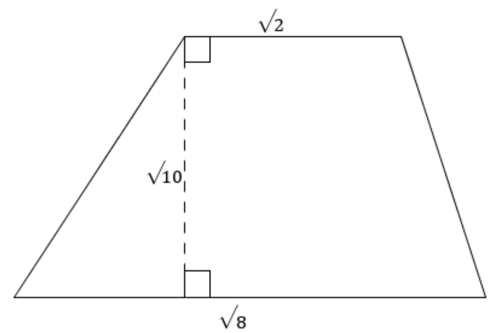
6 Simplify completely:

a $\sqrt{6}+\sqrt{54}$ b $\sqrt{243}+\sqrt{3}$ c $\sqrt{180}+\sqrt{500}$ d $\sqrt{294}-\sqrt{6}$

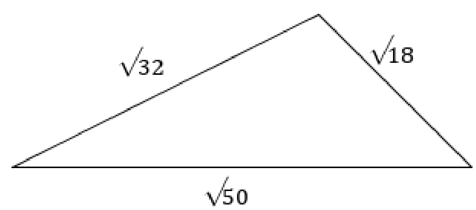
e $\sqrt{32}-\sqrt{98}$ f $4\sqrt{45}+5\sqrt{180}$ g $2\sqrt{75}-3\sqrt{108}$

7 Find the exact area of the rectangle.

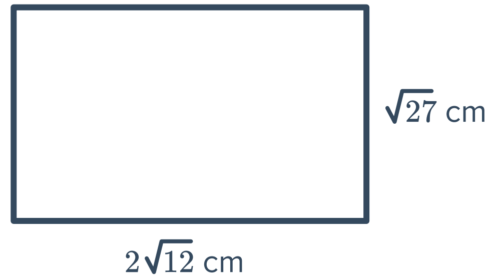
8 Find the area of the trapezoid in simplified radical form.



9 Find the perimeter of the triangle in simplified radical form.



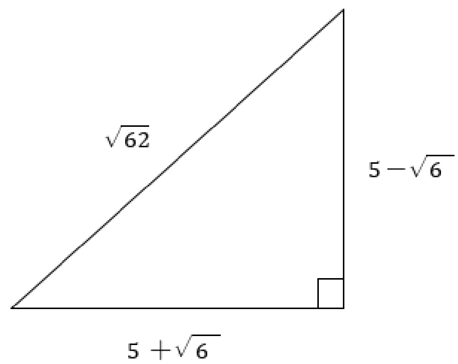
10 Consider the rectangle:



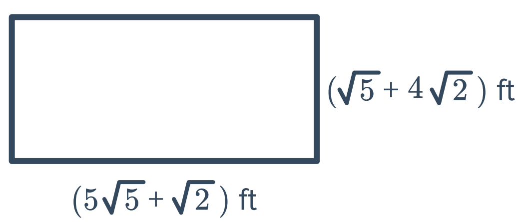
a Find the exact perimeter of the rectangle. Give your answer in the form $a\sqrt{b}$, where a and b are integers.

b Find the exact area of the rectangle.

11 Iain is participating in a race around the track shown in the diagram, which has dimensions in miles. Find the length of one lap around this race track.



12 Bob is about to paint the perimeter of a rectangular handball court, which has the dimensions shown in the diagram.
Find the total perimeter he has to paint.



13 The body surface area of a person in square metres can be modelled by $A=\dfrac{\sqrt{h}\times\sqrt{w}}{56}$, where A is the surface area, h is the height of the person in inches, and w is the weight of the person in pounds.

a Use the model to find the surface area of a person who is 75 inches tall and weighs 172 pounds.

b Hence find the approximate surface area of the person to the nearest hundredth of a square metre.

Additional questions

14 The surface area of a human body can be approximated by the formula

$$A = \frac{\sqrt{h}\sqrt{w}}{60}$$

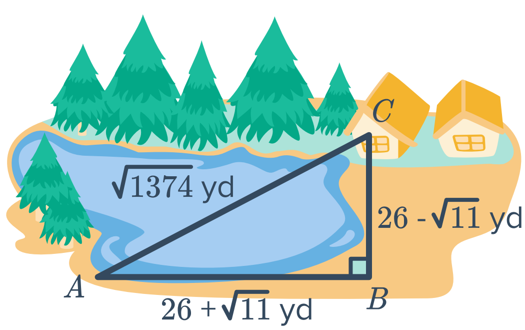
where A is the surface area of the body, h is the height of the person in cm, and w is the weight of the person in kg.

Find the surface area of a person who is 164 cm tall and weighs 63 kg.

15 Uma solved for the height of a triangle whose area is $40\sqrt{65} \text{ ft}^2$ and whose base measures $10\sqrt{13}$ ft.

1. $$\text{Height} = \frac{2\text{Area}}{\text{Base}}$$
2. $$\text{Height} = \frac{2 \cdot 40\sqrt{65}}{10\sqrt{13}}$$
3. $$\text{Height} = \frac{80\sqrt{5 \cdot 13}}{10\sqrt{13}}$$
4. $$\text{Height} = \frac{8\left(\sqrt{5} \cdot \sqrt{13}\right)}{\sqrt{13}}$$
5. $$\text{Height} = 8\sqrt{5}$$

16 Aviva and Jillian are traveling from the shore at Point A to their home at Point C. Jillian wants to swim home and travels directly from A to C. Aviva wants to avoid the water and decides to run home via point B.



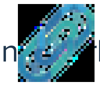
a. How far does Jillian have to swim to get home?

b. How far does Aviva have to run to get home?

c. How much further did Aviva travel than Jillian?

17 Give two different pairs of values for k and m which make the following equation true:

$$\sqrt{k} + \sqrt{m} = \sqrt{k+m}$$

18 Identify and correct the error in the followink:

$$\begin{aligned} 5\sqrt{5} + \sqrt{45} &= \sqrt{125} + \sqrt{45} \\ &= \sqrt{170} \end{aligned}$$

19 Sasha is simplifying the expression $\sqrt[3]{54} - \sqrt[3]{128}$ and says that the answer is $-\sqrt[3]{2}$. Is she correct? Explain how you know.